

Practice SQL Using Oracle XE 11g

Practice 1 (Select Command)

1. Initiate an iSQL*Plus session using the user ID and password provided by the instructor.

```
C:\Users\HP>sqlplus

SQL*Plus: Release 11.2.0.2.0 Production on พท. ก.ย. 26 21:40:14 2024

Copyright (c) 1982, 2014, Oracle. All rights reserved.

Enter user-name: SYSTEM
Enter password:

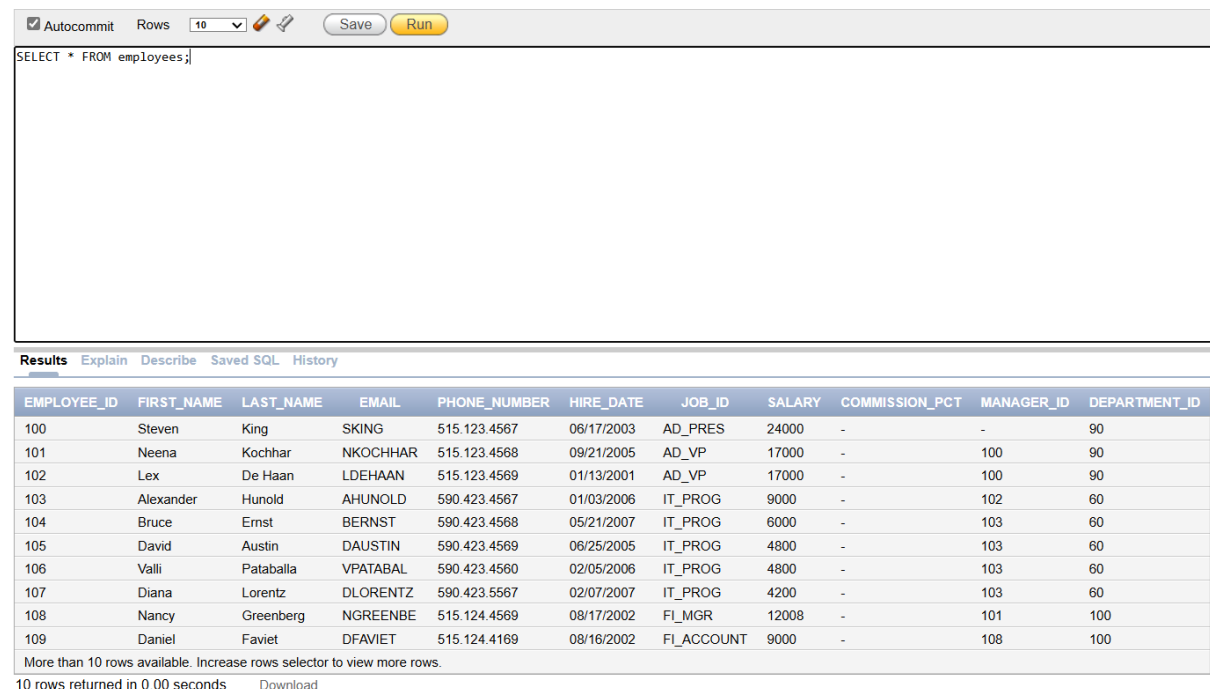
Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production
SQL>
```

- เปิด Command Prompt >> พิมพ์ sqlplus >> Enter

- ใส่ user-name และ password

2. iSQL*Plus commands access the database. True/False

Ans: True จริง



The screenshot shows the iSQL*Plus web interface. At the top, there's a toolbar with 'Autocommit' checked, 'Rows' set to 10, and 'Save' and 'Run' buttons. Below the toolbar, the SQL command 'SELECT * FROM employees;' is entered in the text area. The 'Results' tab is selected, displaying a table with 11 columns: EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, SALARY, COMMISSION_PCT, MANAGER_ID, and DEPARTMENT_ID. The table contains 10 rows of data. At the bottom, a message states 'More than 10 rows available. Increase rows selector to view more rows.' and '10 rows returned in 0.00 seconds' with a 'Download' link.

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	515.123.4567	06/17/2003	AD_PRES	24000	-	-	90
101	Neena	Kochhar	NKOCHHAR	515.123.4568	09/21/2005	AD_VP	17000	-	100	90
102	Lex	De Haan	LDEHAAN	515.123.4569	01/13/2001	AD_VP	17000	-	100	90
103	Alexander	Hunold	AHUNOLD	590.423.4567	01/03/2006	IT_PROG	9000	-	102	60
104	Bruce	Ernst	BERNST	590.423.4568	05/21/2007	IT_PROG	6000	-	103	60
105	David	Austin	DAUSTIN	590.423.4569	06/25/2005	IT_PROG	4800	-	103	60
106	Valli	Pataballa	VPATABAL	590.423.4560	02/05/2006	IT_PROG	4800	-	103	60
107	Diana	Lorentz	DLORENTZ	590.423.5567	02/07/2007	IT_PROG	4200	-	103	60
108	Nancy	Greenberg	NGREENBE	515.124.4569	08/17/2002	FI_MGR	12008	-	101	100
109	Daniel	Faviet	DFAVIET	515.124.4169	08/16/2002	FI_ACCOUNT	9000	-	108	100

3. The following SELECT statement executes successfully:

SELECT last_name, job_id, salary AS Sal FROM employees;

True/False

Ans: True จริง

☒ Autocommit	Rows	10			Save	Run
SELECT last_name, job_id, salary as Sal FROM employees;						
Results Explain Describe Saved SQL History						
LAST_NAME	JOB_ID	SAL				
King	AD_PRES	24000				
Kochhar	AD_VP	17000				
De Haan	AD_VP	17000				
Hunold	IT_PROG	9000				
Ernst	IT_PROG	6000				
Austin	IT_PROG	4800				
Pataballa	IT_PROG	4800				
Lorentz	IT_PROG	4200				
Greenberg	FI_MGR	12008				
Faviet	FI_ACCOUNT	9000				
More than 10 rows available. Increase rows selector to view more rows.						
10 rows returned in 0.00 seconds Download						

4. The following SELECT statement executes successfully:

*SELECT * FROM job_grades;*

True/False

Ans: True จริง

☒ Autocommit	Rows	200			Save	Run
SELECT * FROM job_grades						
Results Explain Describe Saved SQL History						
GRADE_LEVEL	LOWEST_SAL	HIGHEST_SAL				
A	1000	2999				
B	3000	5999				
C	6000	9999				
D	10000	14999				
E	15000	24999				
F	25000	40000				
6 rows returned in 0.00 seconds Download						

5. There are four coding errors in this statement. Can you identify them?

SELECT employee_id, last_name sal x 12 ANNUAL SALARY FROM employees;

Autocommit Rows 10 Save Run

SELECT employee_id, last_name sal x 12 ANNUAL SALARY FROM employees;

Results Explain Describe Saved SQL History

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	515.123.4567	06/17/2003	AD_PRES	24000	-	-	90
101	Neena	Kochhar	NKOCHHAR	515.123.4568	09/21/2005	AD_VP	17000	-	100	90
102	Lex	De Haan	LDEHAAN	515.123.4569	01/13/2001	AD_VP	17000	-	100	90
103	Alexander	Hunold	AHUNOLD	590.423.4567	01/03/2006	IT_PROG	9000	-	102	60
104	Bruce	Ernst	BERNST	590.423.4568	05/21/2007	IT_PROG	6000	-	103	60
105	David	Austin	DAUSTIN	590.423.4569	06/25/2005	IT_PROG	4800	-	103	60
106	Valli	Pataballa	VPATABAL	590.423.4560	02/05/2006	IT_PROG	4800	-	103	60
107	Diana	Lorentz	DLORENTZ	590.423.5567	02/07/2007	IT_PROG	4200	-	103	60
108	Nancy	Greenberg	NGREENBE	515.124.4569	08/17/2002	FI_MGR	12008	-	101	100
109	Daniel	Faviet	DFAVIET	515.124.4169	08/16/2002	FI_ACCOUNT	9000	-	108	100

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds Download

Ans: อธิบาย 4 coding error

1. ไม่ใส่ , ขึ้นระหว่าง last_name กับ sal
2. ไม่มี Colume sal ใน Table employees
3. เครื่องหมายคูณต้องใช้ * แทน x
4. การตั้งชื่อ Column ไม่ควรใช้ space อาจจะใช้ _ แทน หรือได้ “ANNUAL SALARY”

Autocommit Rows 10 Save Run

SELECT employee_id, last_name sal x 12 ANNUAL SALARY FROM employees;
 SELECT employee_id, last_name, salary * 12 AS ANNUAL SALARY FROM employees;
 SELECT employee_id, last_name, salary * 12 AS "ANNUAL SALARY" FROM employees;

Results Explain Describe Saved SQL History

EMPLOYEE_ID	LAST_NAME	ANNUAL SALARY
100	King	288000
101	Kochhar	204000
102	De Haan	204000
103	Hunold	108000
104	Ernst	72000
105	Austin	57600
106	Pataballa	57600
107	Lorentz	50400
108	Greenberg	144096
109	Faviet	108000

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds Download

- แก้ Error แล้ว

6. Show the structure of the DEPARTMENTS table. Select all data from the DEPARTMENTS table.

☒ Autocommit Rows 10 Save Run

SELECT * FROM departments;

Results Explain Describe Saved SQL History

Results

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
30	Purchasing	114	1700
40	Human Resources	203	2400
50	Shipping	121	1500
60	IT	103	1400
70	Public Relations	204	2700
80	Sales	145	2500
90	Executive	100	1700
100	Finance	108	1700

More than 10 rows available. Increase rows selector to view more rows.
10 rows returned in 0.00 seconds [Download](#)

☒ Autocommit Rows 10 Save Run

SELECT * FROM departments;

Results Explain Describe Saved SQL History

Explain

Query Plan

Operation	Options	Object	Rows	Time	Cost	Bytes	Filter Predicates *	Access Predicates
SELECT STATEMENT			27	1	3	567		
TABLE ACCESS	FULL	DEPARTMENTS	27	1	3	567		

* Unindexed columns are shown in red

Index Columns

Owner	Table Name	Index Name	Used In Plan	Columns	Uniqueness	Status	Index Type	Join Index
HR	DEPARTMENTS	DEPT_LOCATION_IX		LOCATION_ID	NONUNIQUE	VALID	NORMAL	NO
		DEPT_ID_FK		DEPARTMENT_ID	UNIQUE	VALID	NORMAL	NO

Table Columns

Table Owner	Table Name	Column Name	Data Type
HR	DEPARTMENTS	DEPARTMENT_ID	NUMBER
		DEPARTMENT_NAME	VARCHAR2
		MANAGER_ID	NUMBER
		LOCATION_ID	NUMBER

Explain

- Query Plan มีทั้งหมด 27 rows

- Index Columns

Primary key = department_id

Foreign key = location_id

- Table Columns มีทั้งหมด 4 Columns และ Data type ดังนี้

1. department_id >> เก็บข้อมูลประเภท NUMBER ตัวเลขจำนวนเต็ม
2. department_name >> เก็บข้อมูลประเภท VARCHAR2 ก็คือ text หรือ string
3. manager_id >> เก็บข้อมูลประเภท NUMBER ตัวเลขจำนวนเต็ม
4. location_id >> เก็บข้อมูลประเภท NUMBER ตัวเลขจำนวนเต็ม

7. Show the structure of the EMPLOYEES table. Create a query to display the last name, job code, hire date, and employee number for each employee, with employee number appearing first. Provide an alias STARTDATE for the HIRE_DATE column. Save your SQL statement to a file named lab1_7.sql.

Ans:

☒ Autocommit
 Rows 10
Save Run

SELECT * FROM employees;

Results Explain Describe Saved SQL History

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	515.123.4567	06/17/2003	AD_PRES	24000	-	-	90
101	Neena	Kochhar	NKOCHHAR	515.123.4568	09/21/2005	AD_VP	17000	-	100	90
102	Lex	De Haan	LDEHAAN	515.123.4569	01/13/2001	AD_VP	17000	-	100	90
103	Alexander	Hunold	AHUNOLD	590.423.4567	01/03/2006	IT_PROG	9000	-	102	60
104	Bruce	Ernst	BERNST	590.423.4568	05/21/2007	IT_PROG	6000	-	103	60
105	David	Austin	DAUSTIN	590.423.4569	06/25/2005	IT_PROG	4800	-	103	60
106	Valli	Pataballa	VPATABAL	590.423.4560	02/05/2006	IT_PROG	4800	-	103	60
107	Diana	Lorentz	DLORENTZ	590.423.5567	02/07/2007	IT_PROG	4200	-	103	60
108	Nancy	Greenberg	NGREENBE	515.124.4569	08/17/2002	FI_MGR	12008	-	101	100
109	Daniel	Faviet	DFAVIET	515.124.4169	08/16/2002	FI_ACCOUNT	9000	-	108	100

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

SELECT * FROM employees;

Results Explain Describe Saved SQL History

Query Plan

Operation	Options	Object	Rows	Time	Cost	Bytes	Filter Predicates *	Access Predicates
SELECT STATEMENT			107	1	3	7,383		
TABLE ACCESS	FULL	EMPLOYEES	107	1	3	7,383		

* Unindexed columns are shown in red

Index Columns

Owner	Table Name	Index Name	Used in Plan	Columns	Uniqueness	Status	Index Type	Join Index
HR	EMPLOYEES	EMP_DEPARTMENT_IX		DEPARTMENT_ID	NONUNIQUE	VALID	NORMAL	NO
		EMP_MANAGER_IX		MANAGER_ID	NONUNIQUE	VALID	NORMAL	NO
		EMP_EMP_ID_PK		EMPLOYEE_ID	UNIQUE	VALID	NORMAL	NO
		EMP_EMAIL_UK		EMAIL	UNIQUE	VALID	NORMAL	NO
		EMP_NAME_IX		LAST_NAME,FIRST_NAME	NONUNIQUE	VALID	NORMAL	NO
		EMP_JOB_IX		JOB_ID	NONUNIQUE	VALID	NORMAL	NO

Table Columns

Table Owner	Table Name	Column Name	Data Type
HR	EMPLOYEES	EMPLOYEE_ID	NUMBER
		FIRST_NAME	VARCHAR2
		LAST_NAME	VARCHAR2
		EMAIL	VARCHAR2
		PHONE_NUMBER	VARCHAR2
		HIRE_DATE	DATE
		JOB_ID	VARCHAR2
		SALARY	NUMBER
		COMMISSION_PCT	NUMBER
		MANAGER_ID	NUMBER
		DEPARTMENT_ID	NUMBER

- Structure of Table employees

☒ Autocommit
 Rows 10
Save Run

SELECT employee_id, last name, job id, hire date AS STARTDATE FROM employees;

Results Explain Describe Saved SQL History

EMPLOYEE_ID	LAST_NAME	JOB_ID	STARTDATE
100	King	AD_PRES	06/17/2003
101	Kochhar	AD_VP	09/21/2005
102	De Haan	AD_VP	01/13/2001
103	Hunold	IT_PROG	01/03/2006
104	Ernst	IT_PROG	05/21/2007
105	Austin	IT_PROG	06/25/2005
106	Pataballa	IT_PROG	02/05/2006
107	Lorentz	IT_PROG	02/07/2007
108	Greenberg	FI_MGR	08/17/2002
109	Faviet	FI_ACCOUNT	08/16/2002

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

Autocommit Rows 10 Save Run

```
SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;
```

Results Explain Describe Saved SQL History

Owner -All Users - Find Rows 10 Go Delete Checked

Owner	Name	Description	SQL	Updated By	Last Updated
SCOTT	lab1_7.sql	-	_id	SCOTT	Now

row(s) 1 - 1 of 1

Save SQL

* Name lab1_7.sql

Description

Cancel Save

- ไปที่ Save >> ตั้งชื่อ lab1_7.sql >> กด Save

8. Run your query in the file lab1_7.sql.

Autocommit Rows 10 Save Run

```
SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;
```

Results Explain Describe Saved SQL History

Owner -All Users - Find Rows 10 Go Delete Checked

Owner	Name	Description	SQL	Updated By	Last Updated
SCOTT	lab1_7.sql	-	SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;	SCOTT	4 minutes ago

row(s) 1 - 1 of 1

- ค้นหาไฟล์ lab1_7.sql จาก Saved SQL และนำ Script มา RUN ตามจอทซ์

Autocommit Rows 10 Save Run

```
SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;
```

Results Explain Describe Saved SQL History

EMPLOYEE_ID	LAST_NAME	JOB_ID	STARTDATE
100	King	AD_PRES	06/17/2003
101	Kochhar	AD_VP	09/21/2005
102	De Haan	AD_VP	01/13/2001
103	Hunold	IT_PROG	01/03/2006
104	Ernst	IT_PROG	05/21/2007
105	Austin	IT_PROG	06/25/2005
106	Pataballa	IT_PROG	02/05/2006
107	Lorentz	IT_PROG	02/07/2007
108	Greenberg	FI_MGR	08/17/2002
109	Faviet	FI_ACCOUNT	08/16/2002

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

9. Create a query to display unique job codes from the EMPLOYEES table.

Autocommit Rows 30 Save Run

```
SELECT DISTINCT(job_id) FROM employees;
```

Results Explain Describe Saved SQL History

JOB_ID
AC_ACCOUNT
AC_MGR
AD_ASST
AD_PRES
AD_VP
FI_ACCOUNT
FI_MGR
HR_REP
IT_PROG
MK_MAN
MK_REP
PR_REP
PU_CLERK
PU_MAN
SA_MAN
SA_REP
SH_CLERK
ST_CLERK
ST_MAN

19 rows returned in 0.00 seconds [Download](#)

10. Copy the statement from *lab1_7.sql* into the iSQL*Plus Edit window. Name the column headings *Emp* #, *Employee*, *Job*, and *Hire Date*, respectively. Run your query again.

Autocommit Rows 10 Save Run

```
SELECT employee_id AS Emp#, last_name AS Employee, job_id AS Job, hire_date AS Hire Date FROM employees;
```

Results Explain Describe Saved SQL History

EMP#	EMPLOYEE	JOB	HIRE_DATE
100	King	AD_PRES	06/17/2003
101	Kochhar	AD_VP	09/21/2005
102	De Haan	AD_VP	01/13/2001
103	Hunold	IT_PROG	01/03/2006
104	Ernst	IT_PROG	05/21/2007
105	Austin	IT_PROG	06/25/2005
106	Pataballa	IT_PROG	02/05/2006
107	Lorentz	IT_PROG	02/07/2007
108	Greenberg	FI_MGR	08/17/2002
109	Faviet	FI_ACCOUNT	08/16/2002

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

11. Display the last name concatenated with the job ID, separated by a comma and space, and name the column *Employee* and *Title*.

☒ Autocommit Rows: 10 Save Run

```
SELECT last_name AS Employee,  
       job_id AS Title,  
       last_name || ',' || job_id AS Employee Title  
FROM employees;
```

Results Explain Describe Saved SQL History

EMPLOYEE	TITLE	EMPLOYEE_TITLE
Abel	SA_REP	Abel, SA_REP
Ande	SA_REP	Ande, SA_REP
Atkinson	ST_CLERK	Atkinson, ST_CLERK
Austin	IT_PROG	Austin, IT_PROG
Baer	PR_REP	Baer, PR_REP
Baida	PU_CLERK	Baida, PU_CLERK
Banda	SA_REP	Banda, SA_REP
Bates	SA_REP	Bates, SA_REP
Bell	SH_CLERK	Bell, SH_CLERK
Bernstein	SA_REP	Bernstein, SA_REP

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

12. Create a query to display all the data from the EMPLOYEES table. Separate each column by a comma. Name the column THE_OUTPUT.

Autocommit Rows 10 Save Run

```

SELECT
  employee_id || ',' || first_name || ',' || last_name || ',' || email || ',' || phone_number || ',' ||
  hire_date || ',' || job_id || ',' || salary || ',' || commission_pct || ',' || manager_id || ',' || department_id AS THE_OUTPUT
FROM employees

```

Results Explain Describe Saved SQL History

THE_OUTPUT
100,Steven,King,SKING,515.123.4567,06/17/2003,AD_PRES,24000,,,90
101,Neena,Kochhar,NKOCHHAR,515.123.4568,09/21/2005,AD_VP,17000,,100,90
102,Lex,De Haan,LDEHAAN,515.123.4569,01/13/2001,AD_VP,17000,,100,90
103,Alexander,Hunold,AHUNOLD,590.423.4567,01/03/2006,IT_PROG,9000,,102,60
104,Bruce,Ernst,BERNST,590.423.4568,05/21/2007,IT_PROG,6000,,103,60
105,David,Austin,DAUSTIN,590.423.4569,06/25/2005,IT_PROG,4800,,103,60
106,Valli,Pataballa,VPATABAL,590.423.4560,02/05/2006,IT_PROG,4800,,103,60
107,Diana,Lorentz,DLORENTZ,590.423.5567,02/07/2007,IT_PROG,4200,,103,60
108,Nancy,Greenberg,NGREENBE,515.124.4569,08/17/2002,FI_MGR,12008,,101,100
109,Daniel,Faviet,DFAVIET,515.124.4169,08/16/2002,FI_ACCOUNT,9000,,108,100

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

Practice 2 (Restricting and Sorting Data)

1. Create a query to display the last name and salary of employees earning more than \$12,000. Place your SQL statement in a text file named *lab2_1.sql*. Run your query.

Autocommit Rows 10 Save Run

```

SELECT last_name, salary
FROM employees
WHERE salary >= 12000;

```

Results Explain Describe Saved SQL History

LAST_NAME	SALARY
King	24000
Kochhar	17000
De Haan	17000
Greenberg	12008
Russell	14000
Partners	13500
Errazuriz	12000
Hartstein	13000
Higgins	12008

9 rows returned in 0.00 seconds [Download](#)

Save SQL

* Name

Description

Cancel Save

Autocommit Rows 10 Save Run

```
SELECT last_name, salary
FROM employees
WHERE salary >= 12000;
```

Results Explain Describe Saved SQL History

Owner - All Users - Find Rows 10 Go Delete Checked

	Owner	Name	Description	SQL	Updated By	Last Updated
☐	SCOTT	lab1_7.sql	-	SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;	SCOTT	8 days ago
☐	SCOTT	lab2_1.sql	-	SELECT last_name, salary FROM employees WHERE salary >= 12000;	SCOTT	Now

row(s) 1 - 2 of 2

2. Create a query to display the employee last name and department number for employee number 176.

Autocommit Rows 10 Save Run

```
SELECT last_name, department_id
FROM employees
WHERE employee_id = 176;
```

Results Explain Describe Saved SQL History

LAST_NAME	DEPARTMENT_ID
Taylor	80

1 rows returned in 0.00 seconds Download

3. Modify *lab2_1.sql* to display the last name and salary for all employees whose salary is not in the range of \$5,000 and \$12,000. Place your SQL statement in a text file named *lab2_3.sql*.

☒ Autocommit
 Rows 10
Save Run

```

SELECT last_name, salary
FROM employees
WHERE salary NOT BETWEEN 5000 AND 12000;
    
```

Results Explain Describe Saved SQL History

LAST_NAME	SALARY
King	24000
Kochhar	17000
De Haan	17000
Austin	4800
Pataballa	4800
Lorentz	4200
Greenberg	12008
Khoo	3100
Baida	2900
Tobias	2800

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

Save SQL

Name lab2_3.sql

Description

Cancel Save

☒ Autocommit
 Rows 10
Save Run

```

SELECT last_name, salary
FROM employees
WHERE salary NOT BETWEEN 5000 AND 12000;
    
```

Results Explain Describe Saved SQL History

Owner - All Users - Find Rows 10 Go Delete Checked

	Owner	Name	Description	SQL	Updated By	Last Updated
☐	SCOTT	lab1_7.sql	-	SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;	SCOTT	8 days ago
☐	SCOTT	lab2_1.sql	-	SELECT last_name, salary FROM employees WHERE salary >= 12000;	SCOTT	10 minutes ago
☐	SCOTT	lab2_3.sql	-	SELECT last_name, salary FROM employees WHERE salary NOT BETWEEN 5000 AND 12000;	SCOTT	Now

row(s) 1 - 3 of 3

4. Display the employee last name, job ID, and start date of employees hired between February 20, 2005, and May 1, 2005. Order the query in ascending order by start date.

☒ Autocommit Rows: 10 Save Run

```

SELECT * FROM employees

SELECT last_name, job_id, hire_date
FROM employees
WHERE hire_date BETWEEN '02/20/2005' AND '05/01/2005'
ORDER BY hire_date ASC;

```

[Results](#) [Explain](#) [Describe](#) [Saved SQL](#) [History](#)

LAST_NAME	JOB_ID	HIRE_DATE
Bull	SH_CLERK	02/20/2005
Everett	SH_CLERK	03/03/2005
Errazuriz	SA_MAN	03/10/2005
Smith	SA_REP	03/10/2005
Ozer	SA_REP	03/11/2005
Hutton	SA_REP	03/19/2005
Bernstein	SA_REP	03/24/2005
Frapp	ST_MAN	04/10/2005

8 rows returned in 0.01 seconds [Download](#)

5. Display the last name and department number of all employees in departments 20 and 50 in alphabetical order by name.

☒ Autocommit Rows: 10 Save Run

```

SELECT * FROM employees

SELECT last_name, department_id
FROM employees
WHERE department_id = 20 OR department_id = 50
ORDER BY last_name ASC;

```

[Results](#) [Explain](#) [Describe](#) [Saved SQL](#) [History](#)

LAST_NAME	DEPARTMENT_ID
Atkinson	50
Bell	50
Bissot	50
Bull	50
Cabrio	50
Chung	50
Davies	50
Dellinger	50
Dilly	50
Everett	50

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

6. Modify *lab2_3.sql* to list the last name and salary of employees who earn between \$5,000 and \$12,000, and are in department 20 or 50. Label the columns Employee and Monthly Salary, respectively. Resave *lab2_3.sql* as *lab2_6.sql*. Run the statement in *lab2_6.sql*.

☒ Autocommit Rows 10 Save Run

```
SELECT last_name AS Employee,
       salary AS "Monthly Salary"
FROM employees
WHERE (salary BETWEEN 5000 AND 12000) AND (department_id IN (20,50))
```

Results Explain Describe Saved SQL History

EMPLOYEE	Monthly Salary
Weiss	8000
Fripp	8200
Kaufling	7900
Vollman	6500
Mourgos	5800
Fay	6000

6 rows returned in 0.00 seconds [Download](#)

Save SQL

* Name

Description

Cancel Save

☒ Autocommit Rows 10 Save Run

```
SELECT last_name AS Employee,
       salary AS "Monthly Salary"
FROM employees
WHERE (salary BETWEEN 5000 AND 12000) AND (department_id IN (20,50))
```

Results Explain Describe Saved SQL History

Owner - All Users - Find Rows 10 Go Delete Checked

☐	Owner	Name	Description	SQL	Updated By	Last Updated
☐	SCOTT	lab1_7.sql	-	SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;	SCOTT	8 days ago
☐	SCOTT	lab2_1.sql	-	SELECT last_name, salary FROM employees WHERE salary >= 12000;	SCOTT	3 hours ago
☐	SCOTT	lab2_3.sql	-	SELECT last_name, salary FROM employees WHERE salary NOT BETWEEN 5000 AND 12000;	SCOTT	3 hours ago
☐	SCOTT	lab2_6.sql	-	SELECT last_name AS Employee, salary AS "Monthly Salary" FROM employees WHERE (salary BETWEEN 5000 AND 12000) AND	SCOTT	Now

row(s) 1 - 4 of 4

7. Display the last name and hire date of every employee who was hired in 1994.

Ans: ใน Column hire_date ไม่มีพนักงานคนไหนที่ถูกจ้างในปี 1994

☒ Autocommit Rows 100   Save Run

```
SELECT last_name, hire_date
FROM employees
WHERE TO_CHAR(hire_date, 'YYYY') = '1994'
```

Results Explain Describe Saved SQL History

no data found

☒ Autocommit Rows 100   Save Run

```
SELECT last_name, hire_date
FROM employees
WHERE TO_CHAR(hire_date, 'YYYY') = '1994'

SELECT MIN(TO_CHAR(hire_date, 'YYYY')) AS MIN,
       MAX(TO_CHAR(hire_date, 'YYYY')) AS MAX
FROM employees
```

Results Explain Describe Saved SQL History

MIN	MAX
2001	2008

1 rows returned in 0.00 seconds [Download](#)

- ปีที่ต่ำที่สุดที่มีการจ้างงานคือปี 2001

8. Display the last name and job title of all employees who do not have a manager.

☒ Autocommit Rows 100   Save Run

```
SELECT last_name, job_id
FROM employees
WHERE manager_id IS NULL
```

Results Explain Describe Saved SQL History

LAST_NAME	JOB_ID
King	AD_PRES

1 rows returned in 0.00 seconds [Download](#)

9. Display the last name, salary, and commission for all employees who earn commissions. Sort data in descending order of salary and commissions.

☒ Autocommit
 Rows 50
Save Run

```

SELECT * FROM employees

SELECT last_name, salary, commission_pct
FROM employees
WHERE commission_pct IS NOT NULL
ORDER BY salary DESC, commission_pct DESC
    
```

Results Explain Describe Saved SQL History

LAST_NAME	SALARY	COMMISSION_PCT
Russell	14000	.4
Partners	13500	.3
Errazuriz	12000	.3
Ozer	11500	.25
Cambrault	11000	.3
Abel	11000	.3
Vishney	10500	.25
Zlotkey	10500	.2
King	10000	.35
Tucker	10000	.3
Bloom	10000	.2
Fox	9600	.2
Sully	9500	.35
Bernstein	9500	.25
Greene	9500	.15
McEwen	9000	.35
Hall	9000	.25
Hutton	8800	.25
Taylor	8600	.2
Livingston	8400	.2
Smith	8000	.3
Olsen	8000	.2
Doran	7500	.3
Cambrault	7500	.2
Smith	7400	.15
Bates	7300	.15
Marvins	7200	.1
Sewall	7000	.25
Tuvault	7000	.15
Grant	7000	.15
Lee	6800	.1
Ande	6400	.1
Banda	6200	.1
Johnson	6200	.1
Kumar	6100	.1

35 rows returned in 0.00 seconds
 [Download](#)

10. Display the last names of all employees where the third letter of the name is an a.

☒ Autocommit Rows: 10 Save Run

```

SELECT last_name
FROM employees
WHERE last_name LIKE '__a%'
  
```

[Results](#) [Explain](#) [Describe](#) [Saved SQL](#) [History](#)

LAST_NAME
Grant
Grant
Whalen

3 rows returned in 0.00 seconds [Download](#)

11. Display the last name of all employees who have an a and an e in their last name.

☒ Autocommit Rows: 10 Save Run

```

SELECT last_name
FROM employees
WHERE last_name LIKE '%a%' OR last_name LIKE '%e%';
  
```

[Results](#) [Explain](#) [Describe](#) [Saved SQL](#) [History](#)

LAST_NAME
Abel
Ande
Baer
Baida
Banda
Bates
Bell
Bernstein
Cabrio
Cambrault
More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

12. Display the last name, job, and salary for all employees whose job is sales representative or stock clerk and whose salary is not equal to \$2,500, \$3,500, or \$7,000.

☒ Autocommit
 Rows 20
Save Run

```

SELECT * FROM employees

SELECT last_name, job_id, salary
FROM employees
WHERE job_id IN ('SA_REP', 'ST_CLERK') AND salary NOT IN (2500,3500,7000)
    
```

Results Explain Describe Saved SQL History

LAST_NAME	JOB_ID	SALARY
Nayer	ST_CLERK	3200
Mikkilineni	ST_CLERK	2700
Landry	ST_CLERK	2400
Markle	ST_CLERK	2200
Bissot	ST_CLERK	3300
Atkinson	ST_CLERK	2800
Olson	ST_CLERK	2100
Mallin	ST_CLERK	3300
Rogers	ST_CLERK	2900
Gee	ST_CLERK	2400
Philtanker	ST_CLERK	2200
Ladwig	ST_CLERK	3600
Stiles	ST_CLERK	3200
Seo	ST_CLERK	2700
Davies	ST_CLERK	3100
Matos	ST_CLERK	2600
Tucker	SA_REP	10000
Bernstein	SA_REP	9500
Hall	SA_REP	9000
Olsen	SA_REP	8000

More than 20 rows available. Increase rows selector to view more rows.

20 rows returned in 0.00 seconds [Download](#)

13. Modify *lab2_6.sql* to display the last name, salary, and commission for all employees whose commission amount is 20%. Resave *lab2_6.sql* as *lab2_13.sql*. Rerun the statement in *lab2_13.sql*.

☒ Autocommit Rows: 20

```

SELECT last_name,
       salary,
       CONCAT((commission_pct * 100), '%') AS commission_pct
FROM employees
WHERE commission_pct = .2

```

[Results](#) [Explain](#) [Describe](#) [Saved SQL](#) [History](#)

LAST_NAME	SALARY	COMMISSION_PCT
Zlotkey	10500	20%
Olsen	8000	20%
Cambraut	7500	20%
Bloom	10000	20%
Fox	9600	20%
Taylor	8600	20%
Livingston	8400	20%

7 rows returned in 0.00 seconds [Download](#)

Save SQL

* Name

lab2_13.sql

Description

☒ Autocommit Rows: 20

```

SELECT last_name,
       salary,
       CONCAT((commission_pct * 100), '%') AS commission_pct
FROM employees
WHERE commission_pct = .2

```

[Results](#) [Explain](#) [Describe](#) [Saved SQL](#) [History](#)

Owner: All Users Find: Rows: 10

☐	Owner	Name	Description	SQL	Updated By	Last Updated
☐	SCOTT	lab1_7.sql	-	SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;	SCOTT	9 days ago
☐	SCOTT	lab2_1.sql	-	SELECT last_name, salary FROM employees WHERE salary >= 12000;	SCOTT	6 hours ago
☐	SCOTT	lab2_3.sql	-	SELECT last_name, salary FROM employees WHERE salary NOT BETWEEN 5000 AND 12000;	SCOTT	6 hours ago
☐	SCOTT	lab2_6.sql	-	SELECT last_name AS Employee, salary AS "Monthly Salary" FROM employees WHERE (salary BETWEEN 5000 AND 12000) AND	SCOTT	3 hours ago
☐	SCOTT	lab2_13.sql	-	SELECT last_name, salary, CONCAT((commission_pct * 100), '%') AS commission_pct FROM employees WHERE commis	SCOTT	Now

row(s) 1 - 5 of 5

Practice 3 (Single row function)

1. Write a query to display the current date. Label the column Date.

The screenshot shows the SQL Developer interface. At the top, there's a toolbar with 'Autocommit' checked, 'Rows' set to 10, and 'Save' and 'Run' buttons. The SQL editor contains two queries: `SELECT SYSDATE AS "Date" FROM dual` and `SELECT CURRENT_DATE AS "Date" FROM dual`. Below the editor, the 'Results' tab is active, showing a single row with the date '10/06/2024'. The status bar indicates '1 rows returned in 0.00 seconds' and a 'Download' link.

```
SELECT SYSDATE AS "Date" FROM dual
```

```
SELECT CURRENT_DATE AS "Date" FROM dual
```

Date
10/06/2024

1 rows returned in 0.00 seconds [Download](#)

2. For each employee, display the employee number, last_name, salary, and salary increased by 15% and expressed as a whole number. Label the column New Salary. Place your SQL statement in a text file named lab3_2.sql.

The screenshot shows the SQL Developer interface. The SQL editor contains the following query: `SELECT employee_id, last_name, salary, ROUND(salary * 1.15, 0) AS "New Salary" FROM employees;`. Below the editor, the 'Results' tab is active, displaying a table with 10 rows of employee data. The status bar indicates '10 rows returned in 0.00 seconds' and a 'Download' link. A 'Save SQL' dialog box is open, showing the file name 'lab3_2.sql' and a 'Save' button.

```
SELECT employee_id, last_name, salary, ROUND(salary * 1.15, 0) AS "New Salary" FROM employees;
```

EMPLOYEE_ID	LAST_NAME	SALARY	New Salary
100	King	24000	27600
101	Kochhar	17000	19550
102	De Haan	17000	19550
103	Hunold	9000	10350
104	Ernst	6000	6900
105	Austin	4800	5520
106	Pataballa	4800	5520
107	Lorentz	4200	4830
108	Greenberg	12008	13809
109	Faviet	9000	10350

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

Save SQL
* Name: lab3_2.sql
Description:
[Cancel](#) [Save](#)

Autocommit Rows 10 Save Run

```
SELECT employee_id,
       last_name,
       salary,
       ROUND(salary * 1.15, 0) AS "New Salary"
FROM employees;
```

Results Explain Describe Saved SQL History

Owner All Users Find Rows 10 Go Delete Checked

Owner	Name	Description	SQL	Updated By	Last Updated
SCOTT	lab1_7.sql	-	SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;	SCOTT	9 days ago
SCOTT	lab2_1.sql	-	SELECT last_name, salary FROM employees WHERE salary >= 12000;	SCOTT	28 hours ago
SCOTT	lab2_3.sql	-	SELECT last_name, salary FROM employees WHERE salary NOT BETWEEN 5000 AND 12000;	SCOTT	28 hours ago
SCOTT	lab2_6.sql	-	SELECT last_name AS Employee, salary AS "Monthly Salary" FROM employees WHERE (salary BETWEEN 5000 AND 12000) AND	SCOTT	25 hours ago
SCOTT	lab2_13.sql	-	SELECT last_name, salary, CONCAT((commission_pct * 100), '%') AS commission_pct FROM employees WHERE commis	SCOTT	22 hours ago
SCOTT	lab3_2.sql	-	SELECT employee_id, last_name, salary, ROUND(salary * 1.15, 0) AS "New Salary" FROM employees;	SCOTT	Now

row(s) 1 - 6 of 6

3. Run your query in the file lab3_2.sql.

Autocommit Rows 10 Save Run

```
SELECT employee_id,
       last_name,
       salary,
       ROUND(salary * 1.15, 0) AS "New Salary"
FROM employees;
```

Results Explain Describe Saved SQL History

EMPLOYEE_ID	LAST_NAME	SALARY	New Salary
100	King	24000	27600
101	Kochhar	17000	19550
102	De Haan	17000	19550
103	Hunold	9000	10350
104	Ernst	6000	6900
105	Austin	4800	5520
106	Pataballa	4800	5520
107	Lorentz	4200	4830
108	Greenberg	12008	13809
109	Faviet	9000	10350

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds Download

4. Modify your query lab3_2.sql to add a column that subtracts the old salary from the new salary. Label the column Increase. Save the contents of the file as lab3_4.sql. Run the revised query.

Autocommit Rows 10 Save Run

```
SELECT employee_id,
       last_name,
       salary,
       ROUND(salary * 1.15, 0) AS "New Salary",
       ROUND(salary * 1.15, 0) - salary AS Increase
FROM employees;
```

Results Explain Describe Saved SQL History

EMPLOYEE_ID	LAST_NAME	SALARY	New Salary	INCREASE
100	King	24000	27600	3600
101	Kochhar	17000	19550	2550
102	De Haan	17000	19550	2550
103	Hunold	9000	10350	1350
104	Ernst	6000	6900	900
105	Austin	4800	5520	720
106	Pataballa	4800	5520	720
107	Lorentz	4200	4830	630
108	Greenberg	12008	13809	1801
109	Faviet	9000	10350	1350

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

Save SQL

* Name lab3_4.sql

Description

Cancel Save

Autocommit Rows 10 Save Run

```
SELECT employee_id,
       last_name,
       salary,
       ROUND(salary * 1.15, 0) AS "New Salary",
       ROUND(salary * 1.15, 0) - salary AS Increase
FROM employees;
```

Results Explain Describe Saved SQL History

Owner: All Users Find Rows 10 Go Delete Checked

Owner	Name	Description	SQL	Updated By	Last Updated
SCOTT	lab1_7.sql	-	SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;	SCOTT	10 days ago
SCOTT	lab2_1.sql	-	SELECT last_name, salary FROM employees WHERE salary >= 12000;	SCOTT	48 hours ago
SCOTT	lab2_3.sql	-	SELECT last_name, salary FROM employees WHERE salary NOT BETWEEN 5000 AND 12000;	SCOTT	48 hours ago
SCOTT	lab2_6.sql	-	SELECT last_name AS Employee, salary AS "Monthly Salary" FROM employees WHERE (salary BETWEEN 5000 AND 12000) AND	SCOTT	45 hours ago
SCOTT	lab2_13.sql	-	SELECT last_name, salary, CONCAT((commission_pct * 100), '%') AS commission_pct FROM employees WHERE commis	SCOTT	42 hours ago
SCOTT	lab3_2.sql	-	SELECT employee_id, last_name, salary, ROUND(salary * 1.15, 0) AS "New Salary" FROM employees;	SCOTT	20 hours ago
SCOTT	lab3_4.sql	-	SELECT employee_id, last_name, salary, ROUND(salary * 1.15, 0) AS "New Salary", ROUND(sala	SCOTT	Now

row(s) 1 - 7 of 7

5. Write a query that displays the employee's last names with the first letter capitalized and all other letters lowercase and the length of the name for all employees whose name starts with *J*, *A*, or *M*. Give each column an appropriate label. Sort the results by the employees' last names.

☒ Autocommit
 Rows 100
Save Run

```

SELECT CONCAT( UPPER(SUBSTR(last_name, 1, 1)), LOWER(SUBSTR(last_name, 2)) ) AS capitalize_name,
       LENGTH(last_name) AS length_name
FROM employees
WHERE last_name LIKE 'J%' OR last_name LIKE 'A%' OR last_name LIKE 'M%'
ORDER BY last_name
    
```

Results Explain Describe Saved SQL History

CAPITALIZE_NAME	LENGTH_NAME
Abel	4
Ande	4
Atkinson	8
Austin	6
Johnson	7
Jones	5
Mallin	6
Markle	6
Marlow	6
Marvins	7
Matos	5
Mavris	6
Mccain	6
Mcewen	6
Mikkilineni	11
Mourgos	7

16 rows returned in 0.00 seconds
 [Download](#)

6. For each employee, display the employee's last name, and calculate the number of months between today and the date the employee was hired. Label the column MONTHS_WORKED. Order your results by the number of months employed. Round the number of months up to the closest whole number.

Note: Your results will differ.

☒ Autocommit Rows: 20   Save Run

```

SELECT last_name,
       ROUND(MONTHS_BETWEEN(CURRENT_DATE, hire_date)) AS MONTHS_WORKED
FROM employees
ORDER BY MONTHS_WORKED

```

Results Explain Describe Saved SQL History

LAST_NAME	MONTHS_WORKED
Ande	198
Kumar	198
Banda	198
Markle	199
Philtanker	200
Zlotkey	200
Marvins	200
Lee	200
Geoni	200
Johnson	201
Grant	201
Perkins	202
Popp	202
Gee	202
Tuvault	203
Mourgos	203
Cambrault	204
Colmenares	206
Sullivan	208
OConnell	208

More than 20 rows available. Increase rows selector to view more rows.

20 rows returned in 0.00 seconds [Download](#)

7. Write a query that produces the following for each employee:

<employee last name> earns <salary> monthly but wants <3 times salary>. Label the column Dream Salaries.

Autocommit Rows 10 Save Run

```
SELECT last_name || ' earns ' || salary || ' monthly but wants ' || salary * 3 AS "Dream Salaries"
FROM employees
```

Results Explain Describe Saved SQL History

Dream Salaries
King earns 24000 monthly but wants 72000
Kochhar earns 17000 monthly but wants 51000
De Haan earns 17000 monthly but wants 51000
Hunold earns 9000 monthly but wants 27000
Ernst earns 6000 monthly but wants 18000
Austin earns 4800 monthly but wants 14400
Pataballa earns 4800 monthly but wants 14400
Lorentz earns 4200 monthly but wants 12600
Greenberg earns 12008 monthly but wants 36024
Faviet earns 9000 monthly but wants 27000
More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

8. Create a query to display the last name and salary for all employees. Format the salary to be 15 characters long, left-padded with \$. Label the column SALARY.

Autocommit Rows 10 Save Run

```
SELECT last_name,
       '$' || LPAD(salary,15,' ') AS "SALARY"
FROM employees
```

Results Explain Describe Saved SQL History

LAST_NAME	SALARY
King	\$ 24000
Kochhar	\$ 17000
De Haan	\$ 17000
Hunold	\$ 9000
Ernst	\$ 6000
Austin	\$ 4800
Pataballa	\$ 4800
Lorentz	\$ 4200
Greenberg	\$ 12008
Faviet	\$ 9000
More than 10 rows available. Increase rows selector to view more rows.	

10 rows returned in 0.00 seconds [Download](#)

9. Display each employee's last name, hire date, and salary review date, which is the first Monday after six months of service. Label the column REVIEW. Format the dates to appear in the format similar to "Monday, the Thirty-First of July, 2000."

Autocommit Rows 10 Save Run

```
SELECT last_name,
       hire_date,
       TO_CHAR(NEXT_DAY(ADD_MONTHS(hire_date,6), 'MONDAY'), 'fmDay, "the" Ddspth "of" Month, YYYY') AS REVIEW
FROM employees
```

Results Explain Describe Saved SQL History

LAST_NAME	HIRE_DATE	REVIEW
King	06/17/2003	Monday, the Twenty-Second of December, 2003
Kochhar	09/21/2005	Monday, the Twenty-Seventh of March, 2006
De Haan	01/13/2001	Monday, the Sixteenth of July, 2001
Hunold	01/03/2006	Monday, the Tenth of July, 2006
Ernst	05/21/2007	Monday, the Twenty-Sixth of November, 2007
Austin	06/25/2005	Monday, the Twenty-Sixth of December, 2005
Pataballa	02/05/2006	Monday, the Seventh of August, 2006
Lorentz	02/07/2007	Monday, the Thirteenth of August, 2007
Greenberg	08/17/2002	Monday, the Twenty-Fourth of February, 2003
Faviet	08/16/2002	Monday, the Seventeenth of February, 2003

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

10. Display the last name, hire date, and day of the week on which the employee started. Label the column DAY. Order the results by the day of the week starting with Monday.

Autocommit Rows 10 Save Run

```
SELECT last_name,
       hire_date,
       TO_CHAR(hire_date, 'DAY') AS DAY
FROM employees
ORDER BY TO_CHAR(hire_date-1, 'd')
```

Results Explain Describe Saved SQL History

LAST_NAME	HIRE_DATE	DAY
Ladwig	07/14/2003	MONDAY
Cambrault	10/15/2007	MONDAY
Mallin	06/14/2004	MONDAY
Ernst	05/21/2007	MONDAY
Greene	03/19/2007	MONDAY
Banda	04/21/2008	MONDAY
Walsh	04/24/2006	MONDAY
Ande	03/24/2008	MONDAY
Vollman	10/10/2005	MONDAY
Kumar	04/21/2008	MONDAY

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

- format 'd'= 1-7 ค่า default ของ 'd' Sunday=1, Monday=2, ...

- ถ้าอยากให้ Monday=1 ต้องใช้ -1 ช่วย >> 2-1 = 1 จากนั้นก็จะ Order by โดยที่ Monday = 1 ได้

11. Create a query that displays the employees' last names and commission amounts. If an employee does not earn commission, put "No Commission." Label the column COMM.

Autocommit Rows 10 Save Run

```

SELECT last_name,
       NVL(TO_CHAR(commission_pct), 'No Commission.') AS COMM
FROM employees

SELECT last_name,
       COALESCE(TO_CHAR(commission_pct), 'No Commission.') AS COMM
FROM employees

```

Results Explain Describe Saved SQL History

LAST_NAME	COMM
King	No Commission.
Kochhar	No Commission.
De Haan	No Commission.
Hunold	No Commission.
Ernst	No Commission.
Austin	No Commission.
Pataballa	No Commission.
Lorentz	No Commission.
Greenberg	No Commission.
Faviet	No Commission.
More than 10 rows available. Increase rows selector to view more rows.	

10 rows returned in 0.00 seconds [Download](#)

12. Create a query that displays the employees' last names and indicates the amounts of their annual salaries with asterisks. Each asterisk signifies a thousand dollars. Sort the data in descending order of salary. Label the column EMPLOYEES_AND_THEIR_SALARIES.

☒ Autocommit Rows: 10

```

SELECT last_name || ' ' || rpad('*', ROUND(salary/1000), '*') EMPLOYEES_AND_THEIR_SALARIES
FROM employees
ORDER BY salary DESC

```

Results Explain Describe Saved SQL History

EMPLOYEES_AND_THEIR_SALARIES
King *****
Kochhar *****
De Haan *****
Russell *****
Partners *****
Hartstein *****
Greenberg *****
Higgins *****
Errazuriz *****
Ozer *****
More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

13. Using the DECODE function, write a query that displays the grade of all employees based on the value of the column JOB_ID, as per the following data:

<i>JOB</i>	<i>GRADE</i>
AD_PRES	A
ST_MAN	B
IT_PROG	C
SA_REP	D
ST_CLERK	E
None of the above	0

☒ Autocommit Rows: 10 Save Run

```

SELECT job_id,
       DECODE(job_id,
              'AD_PRES', 'A',
              'ST_MAN', 'B',
              'IT_PROG', 'C',
              'SA_REP', 'D',
              'ST_CLERK', 'E',
              0
              ) AS GRADE
FROM employees;

```

Results Explain Describe Saved SQL History

JOB_ID	GRADE
AC_ACCOUNT	0
AC_MGR	0
AD_ASST	0
AD_PRES	A
AD_VP	0
AD_VP	0
FI_ACCOUNT	0
FI_ACCOUNT	0
FI_ACCOUNT	0
FI_ACCOUNT	0

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

14. Rewrite the statement in the preceding question using the CASE syntax.

Autocommit Rows 10 Save Run

```

SELECT job_id,
       DECODE(job_id,
              'AD_PRES', 'A',
              'ST_MAN', 'B',
              'IT_PROG', 'C',
              'SA_REP', 'D',
              'ST_CLERK', 'E',
              '0'
              ) AS GRADE
FROM employees;

```

Results Explain Describe Saved SQL History

JOB_ID	GRADE
AC_ACCOUNT	0
AC_MGR	0
AD_ASST	0
AD_PRES	A
AD_VP	0
AD_VP	0
FI_ACCOUNT	0
FI_ACCOUNT	0
FI_ACCOUNT	0
FI_ACCOUNT	0

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

Practice 4 (multiple table)

1. Write a query to display the last name, department number, and department name for all employees.

Autocommit Rows 10 Save Run

```

SELECT employees.last name, departments.department id, departments.department name
FROM employees
JOIN departments
ON employees.department id = departments.department id

```

Results Explain Describe Saved SQL History

LAST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
Whalen	10	Administration
Fay	20	Marketing
Hartstein	20	Marketing
Tobias	30	Purchasing
Colmenares	30	Purchasing
Baida	30	Purchasing
Raphaely	30	Purchasing
Khoo	30	Purchasing
Himuro	30	Purchasing
Mavris	40	Human Resources

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

2. Create a unique listing of all jobs that are in department 80. Include the location of the department in the output.

Autocommit Rows 10 Save Run

```

SELECT DISTINCT e.job_id, d.location_id
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE e.department_id = 80;

```

Results Explain Describe Saved SQL History

JOB_ID	LOCATION_ID
SA_REP	2500
SA_MAN	2500

2 rows returned in 0.01 seconds [Download](#)

3. Write a query to display the employee last name, department name, location ID, and city of all employees who earn a commission.

Autocommit Rows 5000 Save Run

```

SELECT * FROM employees
SELECT * FROM departments
SELECT * FROM locations

SELECT e.last_name,
       d.department_name,
       d.location_id,
       l.city
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN locations l ON d.location_id = l.location_id
WHERE e.commission_pct IS NOT NULL

```

Results Explain Describe Saved SQL History

LAST_NAME	DEPARTMENT_NAME	LOCATION_ID	CITY
Russell	Sales	2500	Oxford
Partners	Sales	2500	Oxford
Errazuriz	Sales	2500	Oxford
Cambrault	Sales	2500	Oxford
Zlotkey	Sales	2500	Oxford
Tucker	Sales	2500	Oxford
Bernstein	Sales	2500	Oxford
Hall	Sales	2500	Oxford
Olsen	Sales	2500	Oxford

4. Display the employee last name and department name for all employees who have an *a* (lowercase) in their last names. Place your SQL statement in a text file named lab4_4.sql.

Autocommit Rows 100 Save Run

```

SELECT e.last_name,
       d.department_name
FROM employees e
JOIN departments d ON e.department_id = d.department_id
WHERE e.last_name LIKE 'a%'

```

Results Explain Describe Saved SQL History

LAST_NAME	DEPARTMENT_NAME
Whalen	Administration
Fay	Marketing
Hartstein	Marketing
Baida	Purchasing
Raphaely	Purchasing
Tobias	Purchasing
Colmenares	Purchasing
Mavris	Human Resources
Davies	Shipping
Cabrio	Shipping

Save SQL

* Name lab4_4.sql

Description

Cancel Save

5. Write a query to display the last name, job, department number, and department name for all employees who work in Toronto.

Autocommit Rows 5000 Save Run

```

SELECT * FROM employees
SELECT * FROM departments
SELECT * FROM locations

SELECT e.last_name,
       e.job_id,
       e.department_id,
       d.department_name
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN locations l ON d.location_id = l.location_id
WHERE l.city = 'Toronto'

```

Results Explain Describe Saved SQL History

LAST_NAME	JOB_ID	DEPARTMENT_ID	DEPARTMENT_NAME
Hartstein	MK_MAN	20	Marketing
Fay	MK_REP	20	Marketing

2 rows returned in 0.00 seconds [Download](#)

6. Display the employee last name and employee number along with their manager's last name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively. Place your SQL statement in a text file named lab4_6.sql.

Autocommit Rows 10 Save Run

```

SELECT e.last_name AS "Employee",
       e.employee_id AS "EMP#",
       m.last_name AS "Manager",
       m.employee_id AS "Mgr#"
FROM employees e
join employees m ON e.manager_id = m.employee_id

```

Results Explain Describe Saved SQL History

Employee	EMP#	Manager	Mgr#
Kumar	173	Cambrault	148
Bates	172	Cambrault	148
Smith	171	Cambrault	148
Fox	170	Cambrault	148
Bloom	169	Cambrault	148
Ozer	168	Cambrault	148
Hunold	103	De Haan	102
Banda	167	Errazuriz	147
Ande	166	Errazuriz	147
Lee	165	Errazuriz	147

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

Save SQL

* Name lab4_6.sql

Description

Cancel Save

7. Modify lab4_6.sql to display all employees including King, who has no manager.

Place your SQL statement in a text file named lab4_7.sql. Run the query in lab4_7.sql

Autocommit Rows 200 Save Run

```

SELECT e.last_name AS "Employee",
       e.employee_id AS "EMP#",
       m.last_name AS "Manager",
       m.employee_id AS "Mgr#"
FROM employees e
LEFT join employees m ON e.manager_id = m.employee_id
ORDER BY "Manager" DESC

```

Results Explain Describe Saved SQL History

Employee	EMP#	Manager	Mgr#
King	100	-	-
Abel	174	Zlotkey	149
Johnson	179	Zlotkey	149
Grant	178	Zlotkey	149
Livingston	177	Zlotkey	149
Taylor	176	Zlotkey	149
Hutton	175	Zlotkey	149
Geoni	183	Weiss	120
Sullivan	182	Weiss	120
Taylor	180	Weiss	120
Markle	128	Weiss	120
Landry	127	Weiss	120
Mikkilineni	126	Weiss	120

Save SQL

* Name lab4_7.sql

Description

Cancel Save

8. Create a query that displays employee last names, department numbers, and all the employees who work in the same department as a given employee. Give each column an appropriate label.

Autocommit Rows 200 Save Run

```

SELECT e1.department_id AS department,
       e1.last_name AS employee,
       e2.last_name AS colleague
FROM employees e1
JOIN employees e2 ON e1.department_id = e2.department_id
WHERE e1.employee_id != e2.employee_id
ORDER BY e1.department_id, e1.last_name, e2.last_name;

```

Results Explain Describe Saved SQL History

DEPARTMENT	EMPLOYEE	COLLEAGUE
20	Fay	Hartstein
20	Hartstein	Fay
30	Baida	Colmenares
30	Baida	Himuro
30	Baida	Khoo
30	Baida	Raphaely
30	Baida	Tobias
30	Colmenares	Baida
30	Colmenares	Himuro

9. Show the structure of the JOB_GRADES table. Create a query that displays the name, job, department name, salary, and grade for all employees.

Autocommit Rows 200 Save Run

```

SELECT * FROM job_grades
SELECT * FROM departments
SELECT * FROM employees

SELECT e.last_name,
       e.job_id,
       d.department_name,
       e.salary,
       j.grade_level
FROM employees e, departments d, job_grades j
WHERE e.department_id = d.department_id
AND e.salary BETWEEN j.lowest_sal AND j.highest_sal;

```

Results Explain Describe Saved SQL History

LAST_NAME	JOB_ID	DEPARTMENT_NAME	SALARY	GRADE_LEVEL
Olson	ST_CLERK	Shipping	2100	A
Markle	ST_CLERK	Shipping	2200	A
Philtanker	ST_CLERK	Shipping	2200	A
Landry	ST_CLERK	Shipping	2400	A
Gee	ST_CLERK	Shipping	2400	A
Colmenares	PU_CLERK	Purchasing	2500	A
Marlow	ST_CLERK	Shipping	2500	A
Patel	ST_CLERK	Shipping	2500	A

10. Create a query to display the name and hire date of any employee hired after employee Davies.

Autocommit Rows 10 Save Run

```

SELECT e.last_name,
       e.hire_date
FROM employees e
JOIN employees davies
ON davies.last_name = 'Davies'
AND davies.hire_date < e.hire_date;

```

Results Explain Describe Saved SQL History

LAST_NAME	HIRE_DATE
Kochhar	09/21/2005
Hunold	01/03/2006
Ernst	05/21/2007
Austin	06/25/2005
Pataballa	02/05/2006
Lorentz	02/07/2007
Chen	09/28/2005
Sciarra	09/30/2005
Urman	03/07/2006
Popp	12/07/2007

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

11. Display the names and hire dates for all employees who were hired before their managers, along with their manager's names and hire dates. Label the columns Employee, Emp Hired, Manager, and Mgr Hired, respectively.

Autocommit Rows 10 Save Run

```

SELECT e.last_name,
       e.hire_date,
       m.last_name,
       m.hire_date
FROM employees e
JOIN employees m
ON e.manager_id = m.employee_id
WHERE e.hire_date < m.hire_date;

```

Results Explain Describe Saved SQL History

LAST_NAME	HIRE_DATE	LAST_NAME	HIRE_DATE
Kaufling	05/01/2003	King	06/17/2003
Raphaely	12/07/2002	King	06/17/2003
De Haan	01/13/2001	King	06/17/2003
Higgins	06/07/2002	Kochhar	09/21/2005
Baer	06/07/2002	Kochhar	09/21/2005
Mavris	06/07/2002	Kochhar	09/21/2005
Whalen	09/17/2003	Kochhar	09/21/2005
Greenberg	08/17/2002	Kochhar	09/21/2005
Austin	06/25/2005	Hunold	01/03/2006
Faviet	08/16/2002	Greenberg	08/17/2002

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

Practice 5 (Group function)

Determine the validity of the following three statements. Circle either True or False.

1. Group functions work across many rows to produce one result. **True/False**

Ans: True จริง

2. Group functions include nulls in calculations. **True/False**

Ans: False ไม่จริง

3. The WHERE clause restricts rows prior to inclusion in a group calculation. **True/False**

Ans: True จริง

4. Display the highest, lowest, sum, and average salary of all employees. Label the columns Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number. Place your SQL statement in a text file named lab5_6.sql.

The screenshot shows a SQL IDE interface. The top toolbar includes 'Autocommit', 'Rows' (set to 10), 'Save', and 'Run' buttons. The query editor contains the following SQL statement:

```
SELECT e.last_name AS "Employee",
       e.employee_id AS "EMP#",
       m.last_name AS "Manager",
       m.employee_id AS "Mgr#"
FROM employees e
join employees m ON e.manager_id = m.employee_id
```

Below the editor is a 'Saved SQL' table with columns: Owner, Name, Description, and SQL. The table lists several saved queries by user SCOTT. A 'Save SQL' dialog box is open, showing the 'Name' field with 'lab4_6.sql' and the 'Description' field empty. The dialog has 'Cancel' and 'Save' buttons.

Owner	Name	Description	SQL
SCOTT	lab1_7.sql	-	SELECT employee_id, last_name, job_id, hire_date AS STARTDATE FROM employees;
SCOTT	lab2_1.sql	-	SELECT last_name, salary FROM employees WHERE salary >= 12000;
SCOTT	lab2_3.sql	-	SELECT last_name, salary FROM employees WHERE salary NOT BETWEEN 5000 AND 12000;
SCOTT	lab2_6.sql	-	SELECT last_name AS Employee, salary AS "Monthly Salary" FROM employees WHERE (salary BETWEEN 5000 AND 12000) AND
SCOTT	lab2_13.sql	-	SELECT last_name, salary, CONCAT((commission_pct * 100), '%') AS commission_pct FROM employees WHERE commis
SCOTT	lab3_2.sql	-	SELECT employee_id, last_name, salary, ROUND(salary * 1.15, 0) AS "New Salary" FROM employees;
SCOTT	lab3_4.sql	-	SELECT employee_id, last_name, salary, ROUND(salary * 1.15, 0) AS "New Salary", ROUND(sala
SCOTT	lab4_4.sql	-	SELECT e.last_name, d.department_name FROM employees e JOIN departments d ON e.department_id = d.department_id WH
SCOTT	lab4_6.sql	-	SELECT e.last_name AS "Employee", e.employee_id AS "EMP#", m.last_name AS "Manager", m.employee_i

5. Modify the query in lab5_4.sql to display the minimum, maximum, sum, and average salary for each job type. Resave lab5_5.sql to lab5_4.sql. Run the statement in lab5_5.sql.

Script: lab5_5.sql

Status: Complete

View: ☒ Detail ☐ Summary

Show: ☒ Statement ☒ Results ☒ Feedback

Go

```

SELECT job_id,
       ROUND(MAX(salary),0) "Maximum",
       ROUND(MIN(salary),0) "Minimum",
       ROUND(SUM(salary),0) "Sum",
       ROUND(AVG(salary),0) "Average"
FROM employees
GROUP BY job_id

```

JOB_ID	Maximum	Minimum	Sum	Average
IT_PROG	9000	4200	28800	5760
AC_MGR	12008	12008	12008	12008
AC_ACCOUNT	8300	8300	8300	8300
ST_MAN	8200	5800	36400	7280
PU_MAN	11000	11000	11000	11000
AD_ASST	4400	4400	4400	4400
AD_VP	17000	17000	34000	17000
SH_CLERK	4200	2500	64300	3215
FI_ACCOUNT	9000	6900	39600	7920
FI_MGR	12008	12008	12008	12008
PU_CLERK	3100	2500	13900	2780
SA_MAN	14000	10500	61000	12200
MK_MAN	13000	13000	13000	13000
PR_REP	10000	10000	10000	10000
AD_PRES	24000	24000	24000	24000
SA_REP	11500	6100	250500	8350
MK_REP	6000	6000	6000	6000
ST_CLERK	3600	2100	55700	2785
HR_REP	6500	6500	6500	6500

19 rows selected. 0.02 seconds

[Download](#)

Run By: SCOTT

Parsing Schema: HR

Script Started: Sunday, October 20, 2024

Elapsed time: 37 seconds ago

Statements Processed: 1

Successful: 1

With Errors: 0

Workspace: HR User: SCOTT

6. Write a query to display the number of people with the same job.

Determine the number of managers without listing them. Label the column Number of Managers. *Hint: Use the MANAGER_ID column to determine the number of managers.*

☒ Autocommit
 Rows

Save Run

```

SELECT job_id,
       COUNT(manager_id) AS "Number of Managers"
FROM employees
GROUP BY job_id
ORDER BY job_id;

```

[Results](#)
[Explain](#)
[Describe](#)
[Saved SQL](#)
[History](#)

JOB_ID	Number of Managers
AC_ACCOUNT	1
AC_MGR	1
AD_ASST	1
AD_PRES	0
AD_VP	2
FI_ACCOUNT	5
FI_MGR	1
HR_REP	1
IT_PROG	5
MK_MAN	1

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds
[Download](#)

7. Write a query that displays the difference between the highest and lowest salaries. Label the column DIFFERENCE.

Autocommit Rows 10 Save Run

```
SELECT MAX(salary) - MIN(salary) DIFFERENCE
FROM employees;
```

Results Explain Describe Saved SQL History

DIFFERENCE
21900

1 rows returned in 0.00 seconds [Download](#)

8. Display the manager number and the salary of the lowest paid employee for that manager. Exclude anyone whose manager is not known. Exclude any groups where the minimum salary is \$6,000 or less. Sort the output in descending order of salary.

Autocommit Rows 10 Save Run

```
SELECT manager_id,
       MIN(salary)
FROM employees
WHERE manager_id IS NOT NULL
GROUP BY manager_id
HAVING MIN(salary) > 6000
ORDER BY MIN(salary) DESC;
```

Results Explain Describe Saved SQL History

MANAGER_ID	MIN(SALARY)
102	9000
205	8300
145	7000
146	7000
108	6900
147	6200
149	6200
148	6100

8 rows returned in 0.00 seconds [Download](#)

9. Write a query to display each department's name, location, number of employees, and the average salary for all employees in that department. Label the columns Name, Location, Number of People, and

Salary, respectively. Round the average salary to two decimal places.

Autocommit Rows 10 Save Run

```

SELECT d.department_name "Name",
       d.location_id "Location",
       COUNT(*) "Number of People",
       ROUND(AVG(salary), 2) "Salary"
FROM employees e, departments d
WHERE e.department_id = d.department_id
GROUP BY d.department_name, d.location_id;

```

Results Explain Describe Saved SQL History

Name	Location	Number of People	Salary
Administration	1700	1	4400
Marketing	1800	2	9500
Sales	2500	34	8955.88
Purchasing	1700	6	4150
Finance	1700	6	8601.33
IT	1400	5	5760
Executive	1700	3	19333.33
Shipping	1500	45	3475.56
Accounting	1700	2	10154
Human Resources	2400	1	6500

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

10. Create a query that will display the total number of employees and, of that total, the number of employees hired in 2001, 2002, 2003, and 2004. Create appropriate column headings.

Autocommit Rows 10 Save Run

```

SELECT COUNT(*) total,
       SUM(DECODE(TO_CHAR(hire_date, 'YYYY'), 2001, 1, 0)) "2001",
       SUM(DECODE(TO_CHAR(hire_date, 'YYYY'), 2002, 1, 0)) "2002",
       SUM(DECODE(TO_CHAR(hire_date, 'YYYY'), 2003, 1, 0)) "2003",
       SUM(DECODE(TO_CHAR(hire_date, 'YYYY'), 2004, 1, 0)) "2004"
FROM employees;

```

Results Explain Describe Saved SQL History

TOTAL	2001	2002	2003	2004
107	1	7	6	10

1 rows returned in 0.00 seconds [Download](#)

11. Create a matrix query to display the job, the salary for that job based on department number, and the total salary for that job, for departments 20, 50, 80, and 90, giving each column an appropriate heading.

☒ Autocommit Rows Save Run

```

SELECT job_id "Job",
       SUM(DECODE(department_id , 20, salary)) "Dept 20",
       SUM(DECODE(department_id , 50, salary)) "Dept 50",
       SUM(DECODE(department_id , 80, salary)) "Dept 80",
       SUM(DECODE(department_id , 90, salary)) "Dept 90",
       SUM(salary) "Total"
FROM employees
GROUP BY job_id;

```

Results Explain Describe Saved SQL History

Job	Dept 20	Dept 50	Dept 80	Dept 90	Total
IT_PROG	-	-	-	-	28800
AC_MGR	-	-	-	-	12008
AC_ACCOUNT	-	-	-	-	8300
ST_MAN	-	36400	-	-	36400
PU_MAN	-	-	-	-	11000
AD_ASST	-	-	-	-	4400
AD_VP	-	-	-	34000	34000
SH_CLERK	-	64300	-	-	64300
FI_ACCOUNT	-	-	-	-	39600
FI_MGR	-	-	-	-	12008

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

Practice 6 (subquery)

1. Write a query to display the last name and hire date of any employee in the same department as Zlotkey. Exclude Zlotkey.

☒ Autocommit Rows Save Run

```

SELECT last_name,
       hire_date
FROM employees
WHERE department_id = (SELECT department_id
                      FROM employees
                      WHERE last_name = 'Zlotkey')
AND last_name != 'Zlotkey';

```

Results Explain Describe Saved SQL History

LAST_NAME	HIRE_DATE
Russell	10/01/2004
Partners	01/05/2005
Errazuriz	03/10/2005
Cambrault	10/15/2007
Tucker	01/30/2005
Bernstein	03/24/2005
Hall	08/20/2005
Olsen	03/30/2006
Cambrault	12/09/2006
Tuvault	11/23/2007

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

2. Create a query to display the employee numbers and last names of all employees who earn more than the average salary. Sort the results in ascending order of salary.

Autocommit Rows 10 Save Run

```

SELECT employee_id,
       last_name
FROM employees
WHERE salary > (SELECT AVG(salary)
                FROM employees)
ORDER BY salary ASC;

```

Results Explain Describe Saved SQL History

EMPLOYEE_ID	LAST_NAME
203	Mavris
123	Vollman
165	Lee
113	Popp
155	Tuvault
161	Sewall
178	Grant
164	Marvins
172	Bates
171	Smith

More than 10 rows available. Increase rows selector to view more rows.
10 rows returned in 0.01 seconds [Download](#)

3. Write a query that displays the employee numbers and last names of all employees who work in a department with any employee whose last name contains a *u*. Place your SQL statement in a text file named lab6_3.sql. Run your query.

Script Name lab6_3.sql

Find & Replace Undo Redo

```

1 SELECT employee_id,
2       last_name
3 FROM employees
4 WHERE department_id IN (SELECT department_id
5                        FROM employees
6                        WHERE last_name like '%u%');
7

```

ORACLE Application Express

Home Application Builder SQL Workshop Team Development Administration

Home > SQL Workshop > SQL Scripts > Results

Script: lab6_3.sql Status: Complete
View: ☒ Detail ☐ Summary Show: ☒ Statement ☒ Results ☒ Feedback [Go](#)

```

SELECT employee_id,
       last_name
FROM employees
WHERE department_id IN (SELECT department_id
                      FROM employees
                      WHERE last_name like '%u%')

```

EMPLOYEE_ID	LAST_NAME
107	Lorentz
106	Pataballa
105	Austin
104	Ernst
103	Hunold
199	Grant
198	OConnell
197	Feeney
196	Walsh
195	Jones
194	McCain
193	Everett

4. Display the last name, department number, and job ID of all employees whose department location ID is 1700.

☒ Autocommit Rows 10   Save Run

```
SELECT last_name,  
       department_id,  
       job_id  
FROM employees  
WHERE department_id IN (SELECT department_id  
                        FROM departments  
                        WHERE location_id = 1700);
```

Results Explain Describe Saved SQL History

LAST_NAME	DEPARTMENT_ID	JOB_ID
King	90	AD_PRES
Kochhar	90	AD_VP
De Haan	90	AD_VP
Greenberg	100	FI_MGR
Faviet	100	FI_ACCOUNT
Chen	100	FI_ACCOUNT
Sciarra	100	FI_ACCOUNT
Urman	100	FI_ACCOUNT
Popp	100	FI_ACCOUNT
Raphaely	30	PU_MAN

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

5. Display the last name and salary of every employee who reports to King.

☒ Autocommit Rows 10   Save Run

```
SELECT last_name,  
       salary  
FROM employees  
WHERE MANAGER_ID = (SELECT employee_id  
                    FROM employees  
                    WHERE last_name = 'King'  
                    AND MANAGER_ID is null  
                    );
```

Results Explain Describe Saved SQL History

LAST_NAME	SALARY
Kochhar	17000
De Haan	17000
Raphaely	11000
Weiss	8000
Fripp	8200
Kaufling	7900
Vollman	6500
Mourgos	5800
Russell	14000
Partners	13500

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.00 seconds [Download](#)

6. Display the department number, last name, and job ID for every employee in the Executive department.

☒ Autocommit Rows Save Run

```
SELECT e.department_id,  
       e.last_name,  
       e.job_id  
FROM employees e  
JOIN departments d ON e.department_id = d.department_id  
WHERE d.department_name = 'Executive';
```

Results Explain Describe Saved SQL History

DEPARTMENT_ID	LAST_NAME	JOB_ID
90	King	AD_PRES
90	Kochhar	AD_VP
90	De Haan	AD_VP

3 rows returned in 0.00 seconds [Download](#)

7. Modify the query in lab6_3.sql to display the employee numbers, last names, and salaries of all employees who earn more than the average salary and who work in a department with any employee with a *u* in their name. Resave lab6_3.sql to lab6_7.sql. Run the statement in lab6_7.sql.

Script: lab6_7.sql Status: Complete
View: ☒ Detail ☐ Summary Show: ☒ Statement ☒ Results ☒ Feedback

```
SELECT employee_id,  
       last_name,  
       salary  
FROM employees  
WHERE salary > (SELECT AVG(salary) AS avg_salary  
                FROM employees)  
              AND department_id IN (SELECT DISTINCT department_id  
                                    FROM employees  
                                    WHERE last_name LIKE '%u%'  
                                )
```

EMPLOYEE_ID	LAST_NAME	SALARY
103	Hunold	9000
123	Vollman	6500
122	Kaufling	7900
121	Fripp	8200
120	Weiss	8000
177	Livingston	8400
176	Taylor	8600
175	Hutton	8800
174	Abel	11000
172	Bates	7300
171	Smith	7400

Practice 9 (Create and Managing Table)

1. Create the DEPT table based on the following table instance chart. Place the syntax in a script called lab9_1.sql, then execute the statement in the script to create the table. Confirm that the table is created.

Column Name	ID	NAME
Key Type		
Nulls/Unique		
FK Table		
FK Column		
Data type	Number	VARCHAR2
Length	7	25

Script Name

```
1 Create table dept (ID Number(7), Name Varchar2(25));
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
```

Script: lab9_1.sql Status: Complete
View: ☒ Detail ☐ Summary Show: ☒ Statement ☒ Results ☒ Feedback

Create table dept (ID Number(7), Name Varchar2(25))

Table created. 0.06 seconds

[Download](#)

Run By	SCOTT
Parsing Schema	HR
Script Started	Sunday, October 20, 2024
	5 seconds ago
Elapsed time	0.06 seconds
Statements Processed	1
Successful	1
With Errors	0

2. Populate the DEPT table with data from the DEPARTMENTS table. Include only columns that you need.

☒ Autocommit Rows: 10 Save Run

```

INSERT INTO dept
SELECT department_id, department_name
FROM departments;

```

[Results](#) [Explain](#) [Describe](#) [Saved SQL](#) [History](#)

27 row(s) inserted.

0.00 seconds

3. Create the EMP table based on the following table instance chart. Place the syntax in a script called lab9_3.sql, and then execute the statement in the script to create the table. Confirm that the table is created.

Column Name	ID	LAST_NAME	FIRST_NAME	DEPT_ID
Key Type				
Nulls/Unique				
FK Table				
FK Column				
Data type	Number	VARCHAR2	VARCHAR2	Number
Length	7	25	25	7

Script Name:

Find & Replace Undo Redo

```

1 CREATE TABLE emp
2 (id NUMBER(7),
3 last_name VARCHAR2(25),
4 first_name VARCHAR2(25),
5 dept_id NUMBER(7));
6
7

```

Script: lab9_3.sql Status: Complete
 View: ☒ Detail ☐ Summary Show: ☒ Statement ☒ Results ☒ Feedback Go

```

CREATE TABLE emp
(id NUMBER(7),
last_name VARCHAR2(25),
first_name VARCHAR2(25),
dept_id NUMBER(7))

```

Table created. 0.00 seconds

[Download](#)

Run By: SCOTT
 Parsing Schema: HR
 Script Started: Sunday, October 20, 2024
 Elapsed time: 4 seconds ago
 Statements Processed: 0.01 seconds
 Successful: 1
 With Errors: 0

4. Modify the EMP table to allow for longer employee last names. Confirm your modification.

☒ Autocommit Rows

```
ALTER TABLE EMP MODIFY LAST_NAME VARCHAR2(50);
DESCRIBE EMP;
```

Results Explain Describe Saved SQL History

Object Type TABLE Object EMP

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP	ID	NUMBER	-	7	0	-	✓	-	-
	LAST_NAME	VARCHAR2	50	-	-	-	✓	-	-
	FIRST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	DEPT_ID	NUMBER	-	7	0	-	✓	-	-
1 - 4									

5. Confirm that both the DEPT and EMP tables are stored in the data dictionary. (Hint: USER_TABLES)

☒ Autocommit Rows

```
SELECT TABLE_NAME
FROM USER_TABLES
WHERE TABLE_NAME IN ('DEPT', 'EMP');
```

Results Explain Describe Saved SQL History

TABLE_NAME
DEPT
EMP

2 rows returned in 0.01 seconds [Download](#)

6. Create the EMPLOYEES2 table based on the structure of the EMPLOYEES table. Include only the EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY, and DEPARTMENT_ID columns. Name the columns in your new table ID, FIRST_NAME, LAST_NAME, SALARY, and DEPT_ID, respectively.

☒ Autocommit Rows: 10 Save Run

```

Create table employees2 as
Select
Employee_ID as ID,
First_Name,
Last_Name,
Salary,
Department_ID as dept_ID
From employees;

```

Results Explain **Describe** Saved SQL History

Object Type: TABLE Object: EMP

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP	ID	NUMBER	-	7	0	-	✓	-	-
	LAST_NAME	VARCHAR2	50	-	-	-	✓	-	-
	FIRST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	DEPT_ID	NUMBER	-	7	0	-	✓	-	-
									1 - 4

7. Drop the EMP table.

☒ Autocommit Rows: 10 Save Run

```

DROP TABLE EMP;

```

Results Explain Describe Saved SQL History

Table dropped.

0.06 seconds

8. Rename the EMPLOYEES2 table to EMP.

☒ Autocommit Rows  

ALTER TABLE EMPLOYEES2 RENAME TO EMP;

Results Explain Describe Saved SQL History

Table altered.

0.02 seconds

9. Add a comment to the DEPT and EMP table definitions describing the tables. Confirm your additions in the data dictionary.

10. Drop the FIRST_NAME column from the EMP table. Confirm your modification by checking the description of the table.

In the EMP table, mark the DEPT_ID column in the EMP table as UNUSED. Confirm your modification by checking the description of the table.

11. Drop all the UNUSED columns from the EMP table. Confirm your modification by checking the description of the table.